



Session 16A: Integration Fundamentals — FTC and u -Substitution

Phase 2 — Classical Techniques | 80 min

Integration is the art of undoing a derivative. You have two tools: the FTC turns an antiderivative into a definite number, and u -substitution reverses the chain rule. This session is about knowing exactly which step to take, in which order.

Prerequisites: 14A (basic derivatives), 13A (limits)

Part A: Antiderivatives — The Reverse Dictionary

The procedure: Given $f(x)$, find $F(x)$ such that $F'(x) = f(x)$. Always check your answer by differentiating.

Example 1: The Power Rule in Reverse

$\frac{d}{dx}[x^n] = nx^{n-1}$. Reverse it: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (for $n \neq -1$).

Procedure:

1. Bump the exponent up by 1: $n \rightarrow n + 1$.
2. Divide by the new exponent: multiply by $\frac{1}{n+1}$.
3. Add $+C$. Never forget $+C$ on an indefinite integral.

$$\begin{aligned}\int x^3 dx &= \frac{x^4}{4} + C. \text{ Check: differentiate } \frac{x^4}{4} \rightarrow x^3. \checkmark \\ \int \sqrt{x} dx &= \int x^{1/2} dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C. \\ \int \frac{1}{x^2} dx &= \int x^{-2} dx = \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C.\end{aligned}$$

Example 2: The Special Case — $\int \frac{1}{x} dx$

The power rule fails at $n = -1$ (would divide by zero). The antiderivative of $\frac{1}{x}$ is:

$$\int \frac{1}{x} dx = \ln |x| + C.$$

Why absolute value? $\ln x$ is only defined for $x > 0$. $\ln |x|$ covers both $x > 0$ and $x < 0$.

Check: $\frac{d}{dx} \ln |x| = \frac{1}{x}$ for all $x \neq 0$.

Example 3: The Antiderivative Dictionary — Memorize These

$f(x)$	$F(x) = \int f(x) dx$
$x^n (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln x + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\frac{1}{1+x^2}$	$\arctan x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\arcsin x + C$

Procedure for combined functions:

1. Split the integral at each $+$ or $-$ sign: $\int (f \pm g) = \int f \pm \int g$.
2. Pull constants outside: $\int cf = c \int f$.
3. Apply the dictionary to each piece.
4. Combine $+C$ (only one $+C$ needed at the end).

$$\int (3x^2 + 2e^x - \frac{1}{x}) dx = 3 \cdot \frac{x^3}{3} + 2e^x - \ln |x| + C = x^3 + 2e^x - \ln |x| + C.$$

Part B: The FTC — Turning Antiderivatives Into Numbers

The procedure: Find ANY antiderivative F of f . Plug in the upper bound. Plug in the lower bound. Subtract.

Example 4: FTC Step by Step

$$\int_a^b f(x) dx = F(b) - F(a), \text{ where } F' = f.$$

Procedure:

1. Forget the bounds temporarily. Find $F(x)$ — any antiderivative of f .
2. Write $[F(x)]_a^b$ (the vertical bar notation).
3. Plug in the upper bound: $F(b)$.
4. Plug in the lower bound: $F(a)$.
5. Subtract: $F(b) - F(a)$.

$$\int_0^3 x^2 dx:$$

1. $F(x) = \frac{x^3}{3}$ (ignore $+C$ — it cancels in the subtraction).
2. $\left[\frac{x^3}{3}\right]_0^3$.
3. $F(3) = \frac{27}{3} = 9$.
4. $F(0) = 0$.
5. $9 - 0 = 9$.

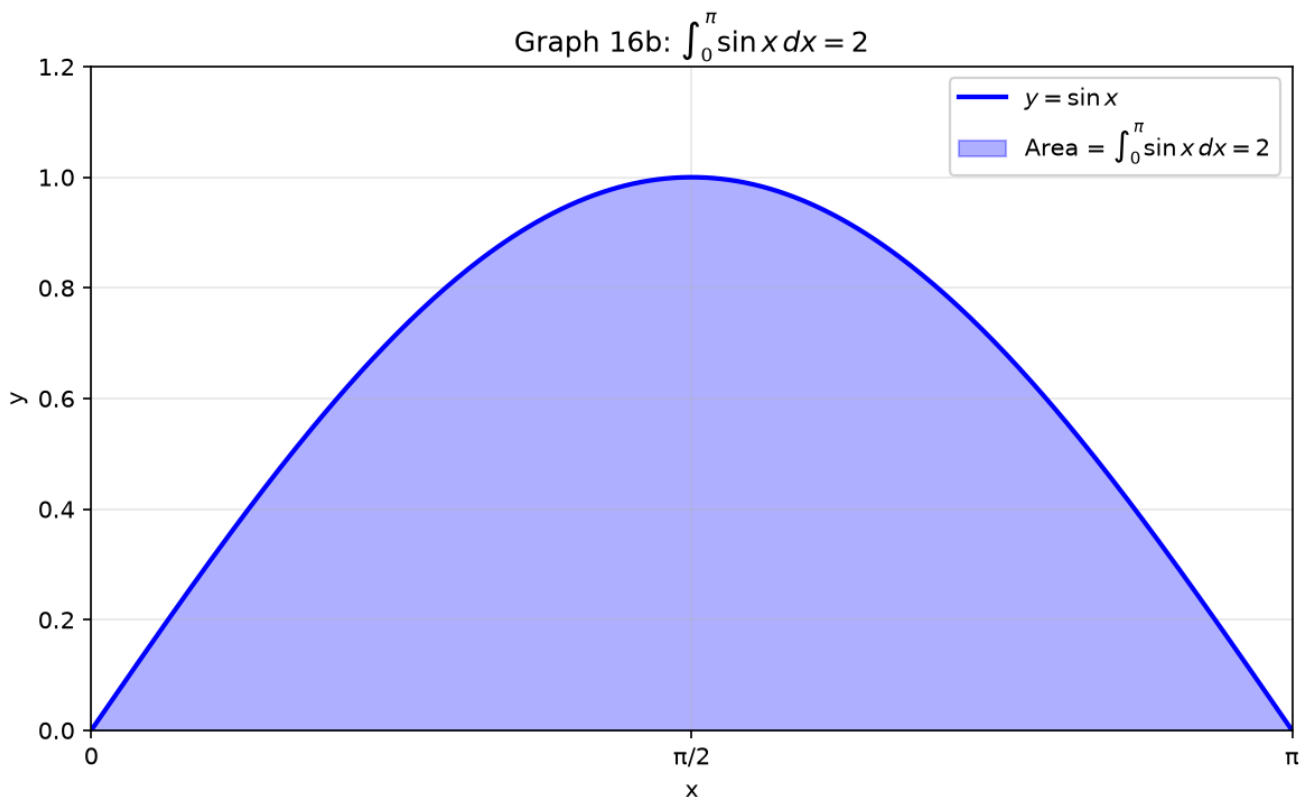
The constant cancels: $\int_0^3 x^2 dx = \left(\frac{27}{3} + C\right) - (0 + C) = 9$. Use the simplest $C = 0$.

Example 5: When the Result Is Zero or Negative

$$\int_0^{\pi} \sin x \, dx = [-\cos x]_0^{\pi} = (-\cos \pi) - (-\cos 0) = (-(-1)) - (-1) = 1 + 1 = 2.$$

$$\int_0^{2\pi} \sin x \, dx = [-\cos x]_0^{2\pi} = (-1) - (-1) = 0. \text{ The positive area cancels the negative.}$$

The definite integral = net signed area. Parts above the x -axis count positive. Parts below count negative.



Example 6: FTC Part 1 — Differentiate an Integral

$$\frac{d}{dx} \int_a^x f(t) \, dt = f(x). \text{ The derivative undoes the integral.}$$

$$\frac{d}{dx} \int_0^x \sin(t^2) \, dt = \sin(x^2). \text{ No need to integrate — just evaluate at } x.$$

With a function in the upper limit (chain rule): $\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x)) \cdot g'(x).$

$$\frac{d}{dx} \int_0^{x^2} \sin t \, dt = \sin(x^2) \cdot 2x.$$

Part C: u -Substitution — Undoing the Chain Rule

The procedure: Spot a function and its derivative side by side. Replace the inner function with u . Replace its derivative dx with du . Integrate in u . Substitute back to x .

Example 7: The u -Sub Algorithm — Five Steps

When to use u -sub: You see a function AND (roughly) its derivative multiplied together.

Algorithm for $\int f(g(x)) \cdot g'(x) \, dx$:

Step	Action	What you write
1	Choose u = the inner function (the one inside parentheses, a root, or a denominator)	$u = g(x)$
2	Compute du = derivative of u , times dx	$du = g'(x) \, dx$
3	Replace all x's — the entire integrand must become a function of u only	$\int (\text{something in } u) \, du$
4	Integrate in u (use the dictionary from Part A)	$F(u) + C$
5	Substitute back $u = g(x)$	$F(g(x)) + C$

Example 8: Polynomial u -Sub — Spot the Derivative

$$\int 2x(x^2 + 1)^5 dx.$$

Step	Action
1	Choose $u = x^2 + 1$ (the inner function, raised to the 5th power)
2	$du = 2x dx$ (exactly the factor multiplying the parentheses!)
3	Replace: $(x^2 + 1)^5 \rightarrow u^5$, and $2x dx \rightarrow du$. Integral becomes $\int u^5 du$
4	Integrate: $\frac{u^6}{6} + C$
5	Back: $\frac{(x^2+1)^6}{6} + C$

Check: Differentiate. $\frac{d}{dx} \left[\frac{(x^2+1)^6}{6} \right] = \frac{6(x^2+1)^5}{6} \cdot 2x = 2x(x^2 + 1)^5. \checkmark$

Example 9: When du Doesn't Match Exactly — Adjust with a Constant

$$\int x\sqrt{x^2 + 4} dx.$$

1. $u = x^2 + 4$.
 2. $du = 2x dx$. But we have $x dx$, not $2x dx$. $\rightarrow$ Solve: $x dx = \frac{1}{2} du$.
 3. Replace: $\sqrt{x^2 + 4} \rightarrow \sqrt{u}$, $x dx \rightarrow \frac{1}{2} du$. Integral: $\int \sqrt{u} \cdot \frac{1}{2} du = \frac{1}{2} \int u^{1/2} du$.
 4. Integrate: $\frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{3} u^{3/2} + C$.
 5. Back: $\frac{1}{3} (x^2 + 4)^{3/2} + C$.
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Example 10: Choosing u — The Decision Rule

Where to look for u (in priority order):

Priority	What to try as u	Example
1st	Expression inside parentheses raised to a power	$(x^2 + 1)^5 \rightarrow u = x^2 + 1$
2nd	Expression inside a root	$\sqrt{x^3 + 1} \rightarrow u = x^3 + 1$
3rd	Denominator of a fraction	$\frac{1}{x^2+1} \rightarrow u = x^2 + 1$
4th	Exponent of e	$e^{x^2} \rightarrow u = x^2$
5th	Inside $\ln$, $\sin$, $\cos$	$\sin(x^2) \rightarrow u = x^2$
6th	The whole base of a power	Try multiple, see what works

$$\int x e^{x^2} dx: \text{Priority 4} \rightarrow u = x^2, du = 2x dx. = \frac{1}{2} e^{x^2} + C.$$

$$\int \frac{\ln x}{x} dx: \text{Priority 5} \rightarrow u = \ln x, du = \frac{1}{x} dx. = \frac{(\ln x)^2}{2} + C.$$

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx: \text{Priority 3 (denominator)} \rightarrow u = \cos x, du = -\sin x dx. = -\ln |\cos x| + C = \ln |\sec x| + C.$$

Example 11: Definite Integrals — Change the Bounds

Procedure for $\int_a^b f(g(x))g'(x) dx$:

1. Set $u = g(x)$. Compute $du = g'(x) dx$.
2. **Change the bounds:** $x = a \rightarrow u = g(a)$, $x = b \rightarrow u = g(b)$.
3. Replace the entire integral: $\int_{g(a)}^{g(b)} (\text{integrand in } u) du$.
4. Integrate in u . Plug in the new bounds. **Do NOT go back to x .**

$$\int_0^1 2x(x^2 + 1)^4 dx.$$

1. $u = x^2 + 1, du = 2x dx$.
2. New bounds: $x = 0 \rightarrow u = 1, x = 1 \rightarrow u = 2$.
3. $\int_1^2 u^4 du$.
4. $[\frac{u^5}{5}]_1^2 = \frac{32}{5} - \frac{1}{5} = \frac{31}{5}$.

Why not go back to x ? You can — the answer is the same. But changing bounds is faster and avoids back-substitution errors.

Example 12: u -Sub Checklist — Did You Do It Right?

After finishing a u -sub, run this checklist:

- ☐ Did every x disappear? (The integrand should have ONLY u and du .)
- ☐ For definite integrals, did you change the bounds?
- ☐ Did you check by differentiating? (Only takes 10 seconds — do it.)
- ☐ Is $+C$ there? (Indefinite integrals only.)

Up to here: Antiderivative dictionary (8 entries). FTC: $\int_a^b f = F(b) - F(a)$. FTC Part 1: $\frac{d}{dx} \int_a^x f = f(x)$. u -sub: 5-step algorithm. Choose u by priority. Definite: change bounds.

Common Mistakes

Mistake 1: Forgetting $+C$ on indefinite integrals

Wrong: $\int x^2 dx = \frac{x^3}{3}$. **Right:** $\int x^2 dx = \frac{x^3}{3} + C$. The $+C$ represents infinitely many possible answers.

Mistake 2: Not changing bounds in definite u -substitution

Wrong: $\int_0^1 2x(x^2 + 1)^4 dx = [\frac{u^5}{5}]_0^1$ with $u = x^2 + 1$. **Right:** Change bounds to u -values:

$$\int_1^2 u^4 du = \left[\frac{u^5}{5}\right]_1^2 = \frac{31}{5}.$$

Mistake 3: $\int \frac{1}{x} dx = \ln x + C$ (missing absolute value)

Wrong: $\ln x$ is undefined for $x < 0$. **Right:** $\ln |x| + C$ works for all $x \neq 0$.

Mistake 4: Trying to integrate a product by integrating each factor separately

Wrong: $\int x \cos x dx = (\int x dx)(\int \cos x dx) = \frac{x^2}{2} \sin x + C$. **Right:** Integration does NOT distribute over multiplication. You need integration by parts (Session 16B).

What We Just Did

- (1) Antiderivative = reverse derivative. Power rule: $\int x^n = x^{n+1}/(n+1) + C$.
Special: $\int 1/x = \ln|x| + C$. Dictionary of 8 standard forms.
- (2) FTC: $\int_a^b f = F(b) - F(a)$. Compute F (any antiderivative), plug bounds, subtract.
FTC Part 1: $d/dx \int_a^x f(t)dt = f(x)$. With chain rule: $f(g(x)) \cdot g'(x)$.
- (3) u-Substitution: 5-step algorithm.
Step 1: Choose u (use priority list).
Step 2: Compute $du = u' \cdot dx$.
Step 3: Replace ALL x 's with u 's.
Step 4: Integrate in u .
Step 5: Substitute back (indefinite) OR change bounds (definite).

Practice 1

$\int (4x^3 - 2x + 5) dx$. Use the antiderivative dictionary — split, pull constants, apply power rule.

→ Solutions: [Solutions](#)

Practice 2

$\int_0^2 (3x^2 + 1) dx$. Apply FTC: find F , compute $F(2) - F(0)$.

→ Solutions: [Solutions](#)

Practice 3

$\int x\sqrt{x^2 + 4} dx$. Run the u -sub algorithm. $u = ?$, $du = ?$, adjust constant?

→ Solutions: [Solutions](#)

Practice 4

$\int_0^{\pi/2} \sin x \cos^2 x dx$. Definite u -sub: choose u , change bounds, integrate, evaluate.

→ Solutions: [Solutions](#)

Practice 5: Real Battle (Constructive)

A student computes $\int_{-2}^2 x^3 dx = \left[\frac{x^4}{4}\right]_{-2}^2 = 4 - 4 = 0$ and concludes "the area under x^3 from -2 to 2 is zero." (a) Is the computation correct? (b) Is the conclusion correct? (c) Compute the TOTAL area (treating all regions as positive) from -2 to 2 . What property of odd functions explains this?

→ Solutions: [Solutions](#)

Basic Algebra Drill — Integration Fundamentals (10 Problems)

Run the prescribed procedure. Show your u , du , and (for definite) new bounds.

- D1.** $\int x^5 dx$. Power rule.
- D2.** $\int (2e^x + \frac{3}{x}) dx$. Dictionary: $e^x \rightarrow e^x$, $\frac{1}{x} \rightarrow \ln|x|$.
- D3.** $\int_1^4 \sqrt{x} dx$. Rewrite as $x^{1/2}$, FTC.
- D4.** $\int_0^\pi \cos x dx$. Dictionary + FTC.
- D5.** $\int 3x^2(x^3 + 1)^4 dx$. u -sub: $u = ?$, $du = ?$ Check: does du match $3x^2 dx$ exactly?
- D6.** $\int e^{3x} dx$. $u = 3x$, $du = 3 dx \rightarrow dx = \frac{du}{3}$.
- D7.** $\int \frac{x}{x^2+1} dx$. $u = x^2 + 1$, $du = 2x dx \rightarrow x dx = \frac{du}{2}$.
- D8.** $\int_0^1 x e^{x^2} dx$. u -sub with bounds.
- D9.** $\int \frac{\cos x}{\sin x} dx$. $u = \sin x$, $du = \cos x dx$. Dictionary for $\frac{1}{u}$.
- D10.** $\int_{-1}^2 (x^2 - 2x) dx$. FTC — evaluate antiderivative at both bounds.

Solutions: [Solutions](#)

Advanced Algebra Drill — Integration Fundamentals (10 Problems)

Multi-step. Choose u . Handle constants. Use FTC creatively.

- A1.** $\int x^2 \sqrt{x^3 + 1} dx$. $u = x^3 + 1$, $du = 3x^2 dx$ — adjust.
- A2.** $\int \frac{e^x}{1+e^{2x}} dx$. $u = e^x$, $du = e^x dx$. Result involves $\arctan(u)$.
- A3.** $\int_0^4 \frac{x}{\sqrt{1+2x}} dx$. $u = 1 + 2x$, solve for x in terms of u , change bounds.

A4. $\int \sin^5 x \cos x \, dx$. $u = \sin x$, $du = \cos x \, dx$ — exact match.

A5. $\int_1^e \frac{(\ln x)^2}{x} \, dx$. $u = \ln x$, change bounds: $x = 1 \rightarrow u = 0$, $x = e \rightarrow u = 1$.

A6. $F(x) = \int_0^x \sin(t^2) \, dt$. Find $F'(x)$. (FTC Part 1 — no integration needed.)

A7. $\int \frac{1}{x \ln x} \, dx$. $u = \ln x$, $du = \frac{1}{x} \, dx$. Result: $\ln |\ln x| + C$.

A8. $\int_0^{\pi/4} \tan x \, dx$. Write as $\frac{\sin x}{\cos x}$, $u = \cos x$, change bounds. Careful with sign.

A9. $\int_0^1 \frac{x}{1+x^4} \, dx$. $u = x^2$, $du = 2x \, dx$. Then $\frac{1}{1+u^2} \rightarrow \arctan u$.

A10. Prove $\int_{-a}^a \sin x \, dx = 0$ for any a without computing. Use the fact that $\sin x$ is odd: $\sin(-x) = -\sin x$. What does this say about the area?

Solutions: [Solutions](#)

Today's Procedure

When you see...	Do this...
A basic function (x^n , e^x , $\sin x$, $\frac{1}{x}$, etc.)	Use the antiderivative dictionary directly
A sum or constant multiple	Split at $+/ -$, pull constants out, apply dictionary
A definite integral $\int_a^b f$	Find F , compute $F(b) - F(a)$
A function AND (roughly) its derivative multiplied	Run the 5-step u -sub algorithm
A definite integral with u -sub	Change bounds to u -values; don't go back to x
$\frac{d}{dx} \int_a^x f(t) \, dt$	FTC Part 1: answer is $f(x)$. With chain rule: $f(g(x)) \cdot g'(x)$

Terminology

What we call it	Math term	Notation
reverse derivative	antiderivative / indefinite integral	$\int f(x) dx = F(x) + C$
the $+C$	constant of integration	$C \in \mathbb{R}$
plug bounds and subtract	Fundamental Theorem of Calculus (FTC)	$\int_a^b f = F(b) - F(a)$
differentiate an integral	FTC Part 1	$\frac{d}{dx} \int_a^x f(t) dt = f(x)$
reverse chain rule	u -substitution	$u = g(x), du = g'(x) dx$
net signed area	definite integral	$\int_a^b f(x) dx$
bounds expressed in u	change of limits	$x = a \rightarrow u = g(a)$